The Space Matters More Than the Loss: JEPA Collapse as a Problem of Structure, Not Optimization

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Abstract

When a JEPA model trains in a free latent space, it can collapse — map all inputs to a single point. Existing solutions address this through the loss function: adding regularizers (VICReg, SIGReg) or a generative decoder (PAN). We show that in settings where domain-relevant coordinates are available, the problem is not the loss but the structure of the space. If the axes of the latent space are defined before training — from physical coordinates, semantic features, or cluster centroids — collapse becomes structurally impossible, and regularization is unnecessary.

Four experiments across different modalities confirm this: a pre-defined latent space structure yields +5% accuracy (vision), 38× lower prediction error (states), 14.8× lower error with 37× fewer parameters (pixels), and +18–20 percentage points in representation entropy (speech). An equal-input control confirms the advantage stems from coordinate fixation, not privileged information access: a free encoder receiving identical input is 7.6× worse than prescribed. SIGReg degrades free representations by 1.9× and has no effect on pre-defined ones — regularization treats what the right space structure prevents. These results reframe the collapse problem in structured domains: not "how do we prevent collapse in a free space," but "why does the encoder define the space at all." The solution lies not in more complex optimization, but in choosing the right space before optimization begins.

1. Introduction

JEPA architectures predict latent representations of masked inputs from visible context, without reconstructing pixels (LeCun, 2022; Assran et al., 2023). This formulation has a structural vulnerability: if the encoder can minimize loss by mapping everything to a single point rather than learning to distinguish inputs, it will. This is the problem of representational collapse.

Two main directions have emerged in the current discussion around this problem:

1. Constrain the space through the loss function: VICReg (Bardes et al., 2022) penalizes axis degeneration, SIGReg (Balestriero & LeCun, 2025) forces the embedding distribution toward an isotropic Gaussian.

2. Supplement prediction with reconstruction: PAN (Ozkara et al., 2025) adds a generative decoder, tying latent dynamics to observable pixels.

Both solutions assume the problem is how the model trains. We propose something different: the problem is where the model trains — in which space. If the axes of the latent space are defined before training — from physical coordinates, semantic features, or cluster centroids — then collapse is structurally impossible, and regularization is unnecessary. We call this approach prescribed axes.

We test this approach on four tasks: speech (clustering anchors), vision (UI screenshots), state prediction (Push-T robot), and pixel prediction (CNN vs. pre-defined structure). Different modalities, different architectures, different metrics — same result.

2. Method

2.1 Problem Setup

Consider a JEPA with an encoder f_θ, a predictor g_ϕ, and a target encoder (typically an EMA copy of f_θ). For input x with context x_c and target x_t, the standard training objective is:

$$\mathbf{L = \| g_\phi(f_\theta(x_c)) - sg(f_\theta(x_t)) \|^2}$$

where sg denotes stop-gradient. The collapse problem arises because the trivial solution $f_\theta(x) = c$ for all x achieves L = 0 when g_ϕ learns the identity mapping. Regularization adds penalty terms to L to prevent this degenerate case.

2.2 Prescribed Axes

We replace the learned encoder f_θ with a fixed, pre-defined mapping $h: X \to \mathbb{R}^d$ that extracts semantically meaningful coordinates. The modified objective becomes:

$$\mathbf{L_p = \| g_\phi(h(x_c)) - h(x_t) \|^2}$$

Three differences from the standard formulation: (1) h has no learnable parameters — it is defined by domain knowledge before training begins; (2) no stop-gradient is needed because h is fixed; (3) no EMA target encoder is needed because the target representation $h(x_t)$ is deterministic.

Collapse is impossible by construction: h is fixed and distinguishes inputs, so any constant prediction $g_\phi(z) = c$ yields $L_p > 0$. The only path to zero loss is perfect prediction. A context-independent predictor can reduce loss by learning the marginal mean $E[h(x_t)]$, but cannot reach zero — and in deterministic environments, the mean is a poor approximation when context is informative.

This guarantee is deliberately minimal: it shows that prescribed axes eliminate collapse, not that they produce good predictions. The empirical question — whether the predictor learns useful structure within the prescribed space — is addressed by Experiments 1–4 and the equal-input control (Section 4.4).

Since h is fixed and has no learnable parameters, EMA and stop-gradient are unnecessary — not as a methodological advantage, but as a direct consequence of the formulation. The structure of the space makes regularization unnecessary.

The prescribed mapping uses domain information unavailable to the free encoder. Experiment 4 tests whether a more powerful encoder can compensate, and the equal-input control (Section 4.4) isolates fixation from information access.

2.3 Instantiations Across Experiments

Vision (Shov-JEPA): h(x) = [P(x), F(x), D(x)] — position, functionality, and depth from the View Hierarchy XML, producing a 3D or 12D feature vector per UI element.

State prediction (LeWM State): h(s) = normalize(s[2:5]) — normalized block coordinates (x, y, θ) from the full 5D state vector. Normalization is min-max to [0,1] using dataset-wide constants: x and y are divided by 512 (workspace range), θ by 2π. These constants are fixed before training and do not depend on the training data distribution.

A second prescribed condition, prescribed_full, uses all five state coordinates [agent_x, agent_y, block_x, block_y, block_angle] with the same normalization. Its worse performance (2.4× vs prescribed_block; see Table 3) suggests that including irrelevant dimensions (agent position) dilutes the prediction signal.

Pixel prediction (LeWM Pixel): h is the same state extraction from ground-truth, while the free baseline uses a 4-layer CNN $f_\theta: \mathbb{R}^{3\times96\times96} \to \mathbb{R}^3$.

Speech (Speech JEPA): $h(x) = \mathrm{softmax}(-\|x - \mu_k\|^2/\tau)$ — projection onto frozen cluster centroids (GMM or k-means, K=512, dim=256) in a fixed codebook space. Cluster centroids were fitted on the same LibriSpeech train-clean-100 split used for JEPA training. This means the prescribed encoder has implicit access to training data statistics through the clustering step. We note this as a potential confound.

2.4 Free Baseline

In all experiments, the free baseline uses a learned encoder f_θ with matching output dimensionality. For state prediction: MLP (5→64→64→3). For pixel prediction: CNN (3→32→64→128→256 channels + MLP → 3). The free encoder is trained end-to-end with the predictor. Where applicable, SIGReg regularization is tested as an ablation on both conditions.

2.5 Shared Components

Predictor architecture is identical across all conditions: an MLP mapping H×2×d inputs to d outputs through two hidden layers of 128 units each, with LayerNorm and GELU activations. Action encoder: MLP (2→32→d) with GELU. Both are trained end-to-end with the encoder (or with the predictor only, when the encoder is fixed).

3. Experiments

We evaluate prescribed axes across four domains with different modalities, architectures, and metrics. All experiments compare prescribed representations against free baselines under matched conditions.

3.1 Summary of Results

Experiment	Modality	Prescribed	Free	Gap	Metric
Speech JEPA	Speech	73.7%	53.2%	+20pp	Entropy (↑)
Shov-JEPA	Vision (UI)	72.5%	67.5%	+5%	Accuracy
LeWM State	States	0.004	0.157	38×	Val loss (↓)
LeWM Pixel	Pixels	0.0009	0.013	14.8×	Val loss (↓)

Table 1: Prescribed vs. free performance across four experiments. Experiments are ordered by increasing evidential depth: from pilot studies (1–2) to controlled experiments with ablation (3–4). Metrics are domain-specific and not directly comparable across experiments.

3.2 Experiment 1: Speech JEPA (Clustering)

We begin with speech: a JEPA architecture with clustering-based anchors, inspired by Ioannides et al. (2026). A 2×2 factorial design crossing clustering method (GMM vs. k-means) with assignment type (soft vs. hard). All four conditions use frozen (prescribed) cluster anchors (K=512, dim=256). Baseline: Pure JEPA without anchors. Data: LibriSpeech train-clean-100, 2000 training steps.

Condition	Entropy (%)	Active / 512	Δ vs Pure JEPA
Pure JEPA	53.2	138	—
GMM + Soft	71.5	128	+18.3pp
GMM + Hard	71.5	128	+18.3pp
KM + Soft	73.7	128	+20.5pp
KM + Hard	73.7	128	+20.5pp

Table 2: Representation quality measured by codebook entropy. Table 1 reports the best result (KM + Soft/Hard, 73.7%).

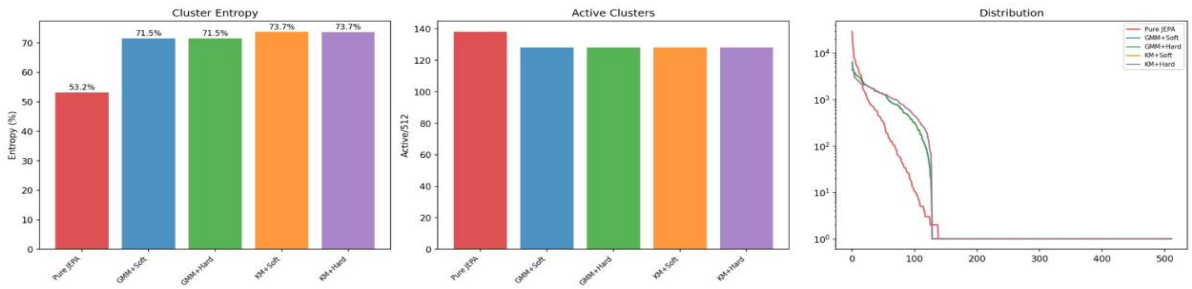

Figure 1: Speech JEPA cluster metrics. All prescribed conditions show substantially higher entropy and more uniform cluster utilization than Pure JEPA.

The factorial analysis reveals a clear hierarchy of effects. The dominant factor is frozen structure itself (+19.4pp averaged across conditions). The clustering method is secondary (k-means +2.2pp over GMM). Assignment type (soft vs. hard) has no measurable effect ($\Delta < 0.03$pp).

This is a pilot study: entropy measures codebook utilization, not downstream representation quality. The pattern, however, is unambiguous — what matters is that the space is prescribed, not how.

3.3 Experiment 2: Shov-JEPA (Vision)

Prescribed axes for UI understanding: Rico dataset (Deka et al., 2017), 398 mobile screenshots. The prescribed representation uses three semantic axes from the View Hierarchy: position (P), functionality (F), depth (D). The free baseline is a 64-dimensional JEPA encoder trained end-to-end.

3 prescribed dimensions yield 72.5%, 64 free dimensions yield 67.5%. The free encoder uses 21× more dimensions — and loses. More dimensions do not compensate for wrong structure — they add capacity where none is needed. A VH-grid JEPA variant combining prescribed features with spatial structure reaches 73.8%, the best result overall. JEPA loss reduction during training: 73–85%, confirming that the predictor learns effectively within the prescribed space.

Unlike Experiment 1, the metric here is classification accuracy — a downstream evaluation: representations trained in prescribed space are used for a separate classification task. This is a pilot study (398 samples, single seed, +5% gap), but it shows the same pattern on a different modality.

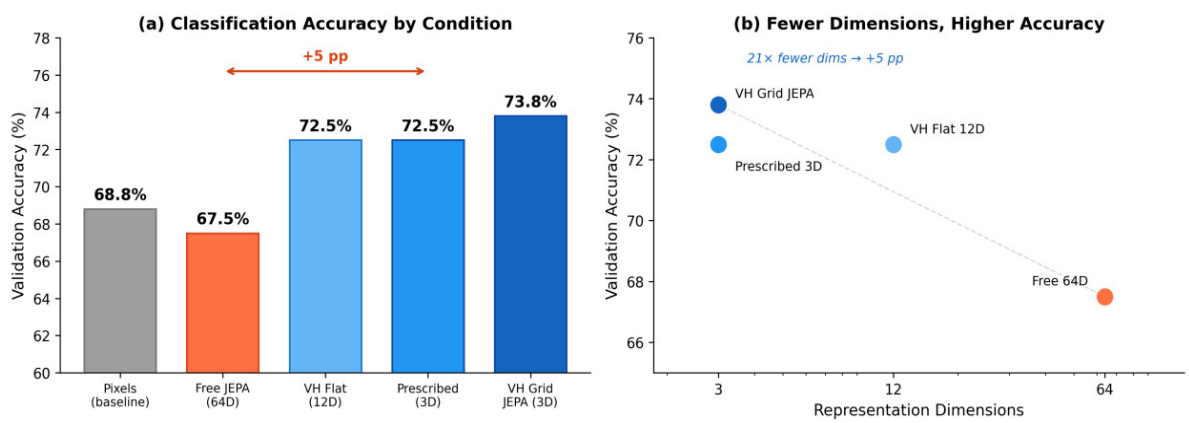

Figure 2: Shov-JEPA results. (a) Classification accuracy across five conditions: prescribed 3D axes achieve 72.5% vs. 67.5% for free 64D encoder (+5 pp). (b) Prescribed axes use 21× fewer dimensions yet achieve higher accuracy.

3.4 Experiment 3: LeWM State (Robotic Control)

Push-T environment (Maes et al., 2026): next-state prediction from history. The prescribed representation uses the block's physical coordinates (x, y, θ), normalized to [0,1]. The free baseline is an MLP encoder (5→64→64→3)

learning its own 3D embedding. Three seeds (42, 123, 777), 50 epochs.

Condition	Seed 42	Seed 123	Seed 777	Mean ± std
Prescribed (block)	0.00366	0.00448	0.00413	0.00409 ± 0.00041
Prescribed (full)	0.00919	0.01147	0.00924	0.00997 ± 0.00130
Free	0.15562	0.15349	0.16170	0.15694 ± 0.00421

Table 3: Validation loss (MSE) across seeds.

For a world model, next-state prediction is not an intermediate metric but the target task: the model exists to predict. A 38× gap means the prescribed world model is qualitatively more accurate than the free one.

The free encoder reaches its best validation loss at epochs 19–22 (depending on seed) and plateaus for the remaining epochs. Prescribed continues improving through epoch 50. This is consistent with representational instability in the free encoder limiting predictor convergence.

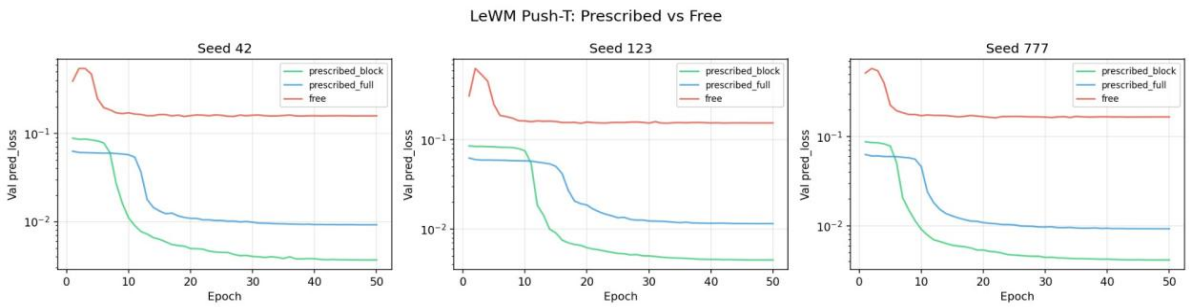

Figure 3: LeWM State training curves across seeds and conditions. Prescribed (green) converges rapidly; free (red) plateaus at high loss.

SIGReg ablation. Removing SIGReg improves the free encoder by 1.9× (mean val loss 0.00599 without vs 0.01161 with, 3 seeds). SIGReg on prescribed: 0.6% effect (0.000469 vs 0.000472). SIGReg and prescribed axes address the same problem (collapse) through different mechanisms. SIGReg constrains what the encoder can do; prescribed axes remove the need to do it. When applied to a space that is already structured, SIGReg has nothing to act on. When applied to a free encoder that has found its own organization, SIGReg disrupts it.

3.5 Experiment 4: LeWM Pixel (CNN Encoder)

One might object: prescribed axes win in Experiment 3 because they receive privileged information — ready-made block coordinates that the free encoder must extract on its own. This is a fair objection. Experiment 4 tests one aspect: whether a more powerful encoder with full visual access can compensate. A direct test — giving the free encoder the same three coordinates — is reported in Section 4.4.

The CNN sees pixels — the entire scene, all objects, all positions. Prescribed axes see three numbers: x, y, θ of the block. CNN: 744K parameters. Prescribed: 20K. Prescribed is 14.8× more accurate (0.000884 vs. 0.01305). The CNN stalls at epoch 7 and shows no improvement across the remaining 43 epochs. Prescribed continues learning through epoch 42.

The CNN uses lr=3e-4, batch size 64, no data augmentation, cosine annealing schedule. No hyperparameter search was performed. A better-tuned CNN or a different architecture (e.g., ResNet) might narrow the gap. The 14.8× result should be read as evidence from a specific configuration, not an optimized comparison.

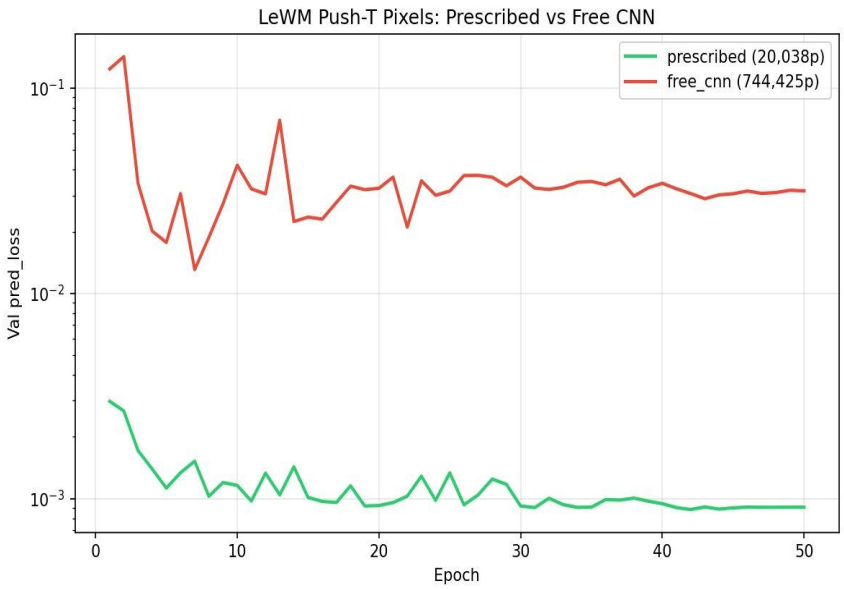

Figure 4: LeWM Pixel training curves. Prescribed (green, 20K params) continues improving through epoch 42; CNN (red, 744K params) plateaus at epoch 7 and never recovers.

The advantage of prescribed here is not privileged data access but separation of concerns. The free encoder solves two tasks simultaneously: learn what to represent and learn how to predict. Prescribed axes remove the first task entirely. 744K parameters split across both tasks lose to 20K parameters focused on one — not because the CNN lacks information, but because it must simultaneously organize the space and work within it.

4. Analysis

4.1 The SIGReg Diagnostic

The SIGReg asymmetry is not just another comparison. It is a diagnostic test. If prescribed axes and SIGReg solve the same problem (collapse), then under prescribed axes SIGReg should change nothing — and it does not (0.6% effect). If under free space SIGReg should help — but it hurts (1.9×), then the free encoder has already found its own spatial organization, and SIGReg breaks it.

Prescribed axes do not conflict with regularization — they make it unnecessary.

4.2 Parameter Efficiency

A free encoder solves two tasks simultaneously: (1) extract useful features and (2) organize the space to prevent collapse. Prescribed axes remove task (2). All capacity goes to (1). This is why 20K prescribed parameters outperform 744K free parameters — most of the free encoder's capacity is spent on a task that prescribed axes solve for free.

4.3 Cross-Modal Consistency

Four modalities, three metrics, different architectures — prescribed wins everywhere. This is not a property of one task. In tasks where meaningful coordinates exist a priori, the structure of the space in which prediction occurs matters more than the power of the mechanism that populates it.

4.4 Equal-Input Control

A natural objection to Experiments 3 and 4 is that prescribed axes receive privileged information: ready-made coordinates that the free encoder must extract from raw state or pixels. To isolate the effect of fixation from information access, we tested a free encoder receiving the same normalized (x, y, θ) input as prescribed — identical information, but with a learned projection (MLP 3→64→64→3) instead of a fixed identity mapping.

Condition	Seed 42	Seed 123	Seed 777	Mean ± std
Prescribed	0.000131	0.000637	0.000646	0.000472 ± 0.000241
Free (5D input)	0.011794	0.011520	0.011527	0.011614 ± 0.000128
Free 3D (same input)	0.000467	0.005024	0.005219	0.003570 ± 0.002195
Free no SIGReg	0.004151	0.006252	0.007558	0.005987 ± 0.001403
Prescribed no SIGReg	0.000135	0.000619	0.000653	0.000469 ± 0.000236

Table 4: Equal-input control and SIGReg ablation. Synthetic Push-T data, 3 seeds, 50 epochs.

Two effects are visible. First, input selection: free_3d (0.00357) is 3.3× better than free_5d (0.01161), confirming that receiving the right coordinates helps even without fixation. Second, fixation: prescribed (0.000472) is 7.6× better than free_3d (0.00357), despite receiving identical information. The advantage of prescribed axes is not from privileged data access but from coordinate fixation.

The free_3d encoder has high variance across seeds (std = 0.00220 vs prescribed std = 0.000241), consistent with the instability of learned representations even when input information is correct.

4.5 Per-Axis Analysis

In the LeWM State experiment, the prescribed advantage is not uniform across axes: position axes (x, y) show 53–63× improvement, the angle axis (θ) shows 25×. Linear quantities are easier to prescribe than cyclical ones. We did not test alternative angle encodings such as (sin θ, cos θ). The 25× gap on the angle axis may partly reflect the linear encoding of a cyclical quantity rather than an intrinsic difficulty. This is a limitation of the current analysis.

5. Related Work

Regularization-based approaches. VICReg (Bardes et al., 2022) adds variance, invariance, and covariance penalties. SIGReg (Balestriero & LeCun, 2025) uses sketched isotropic Gaussian regularization. Both treat collapse as an optimization problem. Our SIGReg experiment (Section 4.1) shows that regularization is not merely unnecessary under pre-defined structure — it is harmful to free encoders.

Architectural approaches. I-JEPA (Assran et al., 2023) uses an EMA target encoder for stable targets. V-JEPA (Bardes et al., 2024) extends this to video. V-JEPA 2.1 (Mur-Labadia et al., 2026) introduces dense predictive loss and hierarchical self-supervision for spatial grounding. These approaches stabilize training but retain the free latent space paradigm.

Generative approaches. PAN (Ozkara et al., 2025) couples latent dynamics with a generative video decoder, arguing that latent prediction without reconstruction is insufficient for grounded world modeling. This represents the second pole of the current debate. Our approach avoids generation entirely by fixing the prediction target space.

Contrastive and non-contrastive approaches. BYOL (Grill et al., 2020) and SimSiam (Chen & He, 2021) prevent collapse through asymmetric architectures without negative pairs. Barlow Twins (Zbontar et al., 2021) decorrelates embedding dimensions. DINO (Caron et al., 2021) uses self-distillation with centering. These methods address collapse more broadly in self-supervised learning. Our work focuses on JEPA architectures; the prescribed axes principle may apply to these frameworks as well, but this remains untested.

Structured representations. Speech JEPA with clustering anchors (Ioannides et al., 2026) is closest to our Experiment 1, using frozen cluster centroids as structured targets. Our factorial analysis extends their work by

isolating the contribution of frozen structure from the clustering method. STP (Huang, LeCun, & Balestriero, 2026) straightens representation trajectories but does not prescribe the space itself.

6. Discussion

6.1 Prescribed Axes as a Reframing of the Problem

The current debate in JEPA research is framed as a choice: regularization (LeCun/Meta FAIR) or generative supervision (PAN/MBZUAI). Both directions answer the same question: “how do we prevent collapse in a free space?” They accept the free latent space as given and look for ways to cope with it.

Prescribed axes do not answer this question. They cancel it. The question prescribed axes ask: “why does the encoder define the space at all?” This is a different level — not a solution method, but a problem formulation.

An analogy from physics: choosing coordinates is not a third method of solving an equation alongside two others. It is a decision made before the choice of method, one that determines how effective any method will be. The analogy has a limit: in physics, coordinate changes are invertible — no information is lost. Prescribed axes are not invertible: h(x) projects into a subspace, and information not encoded in h is discarded. The analogy holds for the principle — a decision made before the method, affecting all downstream performance — but not for information preservation.

The SIGReg diagnostic confirms this empirically. If prescribed axes were an alternative to SIGReg at the same level, SIGReg should either conflict with or complement them. It does neither — it is irrelevant. Prescribed axes operate at a level where regularization is not yet defined.

This claim is falsifiable: it would be refuted if a free encoder, given the same input and sufficient training, consistently matched prescribed performance. Our equal-input control (Section 4.4) shows the opposite: free_3d with identical input is 7.6× worse than prescribed. The claim survives this test.

6.2 Limitations

The main limitation of prescribed axes is the requirement for domain knowledge. In our experiments, the axes are straightforward: physical robot coordinates, semantic UI features, acoustic clusters. But what happens when the right axes are not obvious?

We see three paths. First: prescribed axes from pilot studies — train a small free model, analyze what it learned, and freeze the discovered axes. Second: hierarchical prescribing — define coarse axes (position, scale, type) and leave fine structure free. Third: domain knowledge engineering — in robotics, medicine, and finance, meaningful coordinates are already known to practitioners. None of these paths is tested in this work.

Our experiments are small: 3-dimensional state spaces, hundreds to thousands of samples. We do not know whether the effect holds at 128 or 768 dimensions. In high-dimensional spaces, random projections preserve distances (Johnson–Lindenstrauss), and gauge freedom may be less catastrophic. The scaling question is open.

We compare prescribed axes against free MLP and CNN baselines, not against established self-supervised methods (I-JEPA, V-JEPA, BYOL, DINO). Our free baselines may be weaker than state-of-the-art. The comparison isolates the structural effect (fixed vs learned space), not benchmark performance. Whether prescribed axes improve upon strong free methods is an open question.

Prescribed JEPA is deliberately closer to supervised regression with hand-crafted features than to self-supervised representation learning. We do not claim novelty in the method. The contribution is diagnostic: when we remove the free encoder and replace it with fixed coordinates, system performance improves dramatically. This tells us something about the cost of free encoding, not about the sophistication of prescribed encoding.

All experiments use deterministic environments (or deterministic mappings from state to representation). In stochastic environments, the predictor must approximate a conditional distribution rather than a point, and the advantage of prescribed axes may change. This has not been tested.

Additional confounds we did not isolate: the action encoder must learn in a drifting space for free conditions; weight decay (1e-3) applies differently when the encoder has no learnable parameters; gradient clipping budget is shared differently; the SIGReg implementation generates random projections on each forward pass without fixing the seed. We used the same cosine annealing schedule for all conditions; the free encoder may benefit from alternative schedules.

A fifth experiment is currently underway on the EB-JEPA codebase (Bar et al., 2025), testing whether the same structural effect reproduces on Meta FAIR's architecture with planning success rate as the metric. This is the subject of a separate study.

6.3 Implications

Representation learning assumes by default that representations should be unconstrained — let the model find its own structure. Our results challenge this assumption in domains where meaningful coordinates exist. A free encoder with 37× more capacity loses to prescribed by 14.8×. An equal-input free encoder with identical information loses by 7.6×.

This gap is not due to model weakness. It arises because the free encoder must simultaneously organize the representation space and learn to predict within it — two tasks that prescribed axes separate.

7. Conclusion

Prescribed axes stabilize the latent space across four modalities — vision, states, pixels, speech — and under different metrics: accuracy, prediction loss, entropy. Regularization that is necessary for free spaces becomes unnecessary — and potentially harmful — under prescribed ones. An equal-input control confirms the advantage is from coordinate fixation (7.6×), not information access.

Prescribed axes are not a universal solution. But they show that in structured domains, collapse is a problem of structure, not optimization. And that the solution may lie not in making the loss more complex, but in making the task simpler.

8. Reproducibility

All four experiments are implemented as self-contained notebooks (Google Colab compatible) with preserved execution outputs. Code, data, and results are publicly available at https://github.com/revenue7-eng/prescribed-axes. Experiment 3 includes a standalone Python script with a synthetic data mode for dependency-free verification of the training pipeline. The standalone script's synthetic mode uses simplified physics and does not reproduce the exact numbers reported in the paper, which were obtained using gym-pusht. The equal-input control and SIGReg ablation are implemented in reviewer_response_experiments.py with full result archiving.

Experiment 3 (LeWM State) was run with three seeds; Experiments 1, 2, and 4 with a single seed. For Experiment 2 (Shov-JEPA), a +5% gap with a single seed represents pilot-level evidence. For Experiment 4 (LeWM Pixel), a 14.8× gap makes single-seed less critical, but additional seeds would strengthen the result.

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